

1. Administrative & Study Identification

Full Study Title: A Clinical Study of Calderasib (MK-1084) and Other Treatments for Participants With Non-Small Cell Lung Cancer (MK-1084-007/KANDLELIT-007) **Short Title/Acronym:** KANDLELIT-007 (MK-1084-007) **Protocol Number:** MK-1084-007 **NCT Number:** NCT07190248 **Sponsor Name:** Merck Sharp & Dohme LLC **Investigational Product:** Calderasib (MK-1084) and Subcutaneous Pembrolizumab / Berahyaluronidase alfa (KEYTRUDA QLEX / MK-3475A) **Phase of Development:** Phase 3 **Medical Monitor:** Merck Sharp & Dohme LLC Clinical Director

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate whether calderasib given orally once daily in combination with subcutaneous KEYTRUDA QLEX improves Progression-Free Survival (PFS) in participants with Programmed Death-Ligand 1 (PD-L1) Tumor Proportion Score (TPS) $\geq 1\%$. **Secondary Objectives:** To evaluate PFS in all participants, Overall Survival (OS) in participants with PD-L1 TPS $\geq 1\%$ and in all participants, and Overall Response Rate (ORR).

2.2 Background & Rationale KRAS is the most frequently mutated oncogene in cancer, with the KRAS G12C mutation occurring in approximately 4% to 14% of all patients with lung cancer globally. By pairing the investigational oral calderasib (a KRAS G12C inhibitor) with subcutaneously administered KEYTRUDA QLEX, this study aims to determine whether this chemotherapy-free combination (requiring no intravenous access) can improve survival outcomes compared to subcutaneous pembrolizumab paired with intravenous chemotherapy in newly diagnosed patients.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Trial Status:** Recruiting **Control Type:** Active Comparator **Blinding:** Open-label (Unblinded) **Randomization:** Randomized

3.2 Planned Enrollment & Duration Target N: 675 participants **Number of Sites:** ~116 locations globally **Estimated Study Start:** 08-Oct-2025 **Estimated Primary Completion:** 07-Dec-2029

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Has nonsquamous NSCLC (Stage IIIB, Stage IIIC) not eligible for curative resection or chemoradiation or Stage IV (M1a, M1b, or M1c). 2. Has KRAS G12C-mutant advanced or metastatic nonsquamous NSCLC and has not had prior systemic treatment for advanced disease (first-line treatment). 3. If human immunodeficiency virus (HIV) positive, must have well-controlled HIV on antiretroviral therapy (ART).

4.2 Exclusion Criteria 1. Has diagnosis of small cell lung cancer or, for mixed tumors, presence of small cell elements. 2. Has active inflammatory bowel disease requiring immunosuppressive medication or a previous clear history of inflammatory bowel disease (e.g., Crohn's disease, ulcerative colitis, or chronic diarrhea). 3. Has uncontrolled, significant cardiovascular disease or cerebrovascular disease. 4. Has a gastrointestinal disorder affecting absorption. 5. Is HIV positive and has a history of Kaposi's sarcoma and/or Multicentric Castleman's Disease. 6. Has received prior systemic anticancer therapy for their advanced or metastatic NSCLC. 7. Has known additional malignancy that is progressing or has required active treatment within the past 3 years. 8. Has known active central nervous system (CNS) metastases and/or carcinomatous meningitis. 9. Has active autoimmune disease that has required systemic treatment in the past 2 years. 10. History of (noninfectious) pneumonitis/interstitial lung disease that required steroids or has current pneumonitis/interstitial lung disease. 11. Has active infection requiring systemic therapy except those specified by protocol. 12. Has history of stem cell/solid organ transplant. 13. Has not adequately recovered from major surgery or has ongoing surgical complications.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Calderasib is given orally once daily until discontinuation criteria are met. KEYTRUDA QLEX (MK-3475A) is administered at a fixed dose on Day 1 of each 6-week cycle for up to 18 cycles. **Route of Administration:** Oral (Calderasib) and Subcutaneous (KEYTRUDA QLEX). **Duration of Treatment:** Calderasib until disease progression/discontinuation; MK-3475A for up to 18 cycles (approximately 2 years).

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care measures and medications as permitted by the trial protocol. **Prohibited:** Prior systemic anticancer therapy for advanced/metastatic NSCLC, immunosuppressive medications for active autoimmune diseases within the past 2 years, and systemic steroids for interstitial lung disease.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Tumor Biopsy/KRAS mutation testing, Baseline Scans
Baseline	Day 1	Randomization, First Dose of Study Treatments
Interim	Every 3 to 6 Weeks	Safety Labs, Efficacy Assessment, Tumor Imaging (RECIST v1.1), Study Drug Dispensation
End of Study	Follow-up	Final Vital Signs, Exit Interview, Survival Follow-up Phase

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Progression Free Survival (PFS) in Participants with Programmed Death-Ligand 1 (PD-L1) Tumor Proportion Score (TPS) $\geq 1\%$. PFS is defined as the time from randomization until either documented disease progression per Response Criteria in Solid Tumors Version 1.1 (RECIST 1.1) or death due to any cause, whichever occurs first. (Timeframe: Up to approximately 48 months). **Measurement Method:** Radiographic imaging evaluated per Response Evaluation Criteria in Solid Tumors (RECIST) v1.1. PFS as determined by blinded independent central review (BICR) will be presented.

7.2 Secondary & Exploratory Endpoints * Progression Free Survival (PFS) in All Participants. * Overall Survival (OS) in Participants with PD-L1 TPS $\geq 1\%$. * Overall Survival (OS) in All Participants. * Overall Response Rate (ORR) in Participants with PD-L1 TPS $\geq 1\%$ and in All Participants.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection via site investigator assessments and patient interviews at regular dosing intervals. **Serious Adverse Events (SAEs):** Continuously tracked and reported within 24 hours of site awareness. **Data Monitoring Committee (DMC):** An independent Data Monitoring Committee is utilized for this study.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy analyses (PFS, OS). **Sample Size Justification:** Target enrollment of $N = 675$ is statistically powered to detect meaningful shifts in PFS and OS between the combination therapy arm and the active chemotherapy comparator arm. **Handling of Missing Data:** Standard survival analysis methodologies including censoring at the last evaluable tumor assessment.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.
Monitoring Plan: Standard rigorous onsite and remote monitoring per Sponsor (Merck) guidelines. **Audit Trail:** Utilizing secure eCRF locking and electronic data capture systems with robust data clarification protocols.

11. Study Identifiers & Governance

Primary ID: {{1084-007}} **Registry Number:** {{NCT07190248}}
Sponsor/Lead Organization: {{Merck Sharp & Dohme LLC}} **Study Title:** {{KANDLELIT-007}} **Official Regulatory Title:** {{A Phase 3, Randomized, Open-label, Multicenter Clinical Study to Evaluate the Safety and Efficacy of MK-1084 in Combination With Subcutaneous Pembrolizumab and Berahyaluronidase Alfa (MK-3475A) Versus MK-3475A in Combination With Pemetrexed/Platinum (Carboplatin or Cisplatin) Chemotherapy as First-line Treatment of Participants With KRAS G12C-Mutant, Advanced or Metastatic Nonsquamous NSCLC (KANDLELIT-007)}}

12. Clinical Framework (NSCLC Specific)

Primary Condition: {{Non-Small Cell Lung Cancer (NSCLC)}}
Diagnosis Threshold: {{KRAS G12C-Mutant, Stage IIIB to IV Nonsquamous NSCLC}} **Medical Methodology:** {{Targeted Therapy + Subcutaneous Immunotherapy}} **Optimization Target:** {{PFS and OS prolongation}}

13. Study Procedures & Methodology

Management Pathway: {{Standard clinical pathway for advanced NSCLC}} **Education Protocol (HCP):** {{Study-specific training & subcutaneous administration protocols}} **Education Protocol (Patient):** {{Onsite trial adherence and safety reporting training}} **Quality Audit Mechanism:** {{Clinical site monitoring & Central imaging review}}

14. Observation & Follow-Up Schedule

Total Duration: {{Up to ~8 years (Estimated Study Completion Aug 2032)}} **Onsite Visit Cadence:** {{Every 3 to 6 weeks per treatment cycles}} **Remote Monitoring Frequency:** {{As specified in clinical monitoring plan}} **Checklist Review Interval:** {{Per cycle assessments (every 6 weeks)}}

15. Sign-off & Approval

Role	Name	Signature	Date		:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]					